

DELAWARE CITY, DE, USA

WMO: 997694

Lat: 39.582N

Lon: 75.589W

Elev: 26

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 14.2	(c) 18.4	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 24.9	(k) 29.5	(l) 23.4	(m) 29.6	(n) 9.6	(o) 300	(p) 0.430

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 14.3	(c) 90.8	(d) N/A	(e) 88.4	(f) N/A	(g) 86.2	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 5.4	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 21.8	(b) 18.8	(c) 16.7	(e) 9.4	(f) 95.4	(g) 4.5	(h) 3.0	(i) 6.2	(j) 97.5	(k) 3.6	(l) 99.3	(m) 1.1	(n) 100.9	(o) -2.2	(p) 103.1		
			DB													
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.8	33.8	35.7	43.5	55.9	65.9	74.0	78.2	75.8	69.9	59.3	47.7	40.7
	DBStd	16.99	9.27	8.80	8.42	7.69	6.99	5.36	4.30	3.95	5.63	7.33	7.24	8.10
	HDD50	1619	506	408	235	26	0	0	0	0	0	11	127	306
	HDD65	4318	966	820	666	290	77	3	0	0	17	207	520	752
	CDD50	4109	4	9	34	204	492	720	874	799	598	298	57	20
	CDD65	1331	0	0	0	17	103	274	409	334	165	29	0	0
	CDH74	9678	0	1	4	164	640	1974	3636	2307	839	113	0	0
	CDH80	2859	0	0	0	45	150	587	1281	592	188	16	0	0
(14) Wind	WSAvg	6.9	7.4	7.8	8.5	8.0	6.6	6.3	5.4	5.5	6.3	6.9	7.1	7.2
(15) Precipitation	PrecAvg	41.3	3.0	2.8	3.8	3.3	3.9	3.7	4.1	3.4	4.0	2.8	3.0	3.4
	PrecMax	55.3	8.4	7.0	9.2	6.8	7.4	9.9	12.6	11.2	12.7	6.4	7.8	7.9
	PrecMin	24.9	0.5	0.4	0.8	0.4	0.2	0.2	0.9	0.2	0.8	0.1	0.5	0.8
	PrecStd	7.3	1.6	1.5	1.9	1.6	1.8	1.9	2.3	2.1	2.6	1.5	1.9	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.7	68.0	73.0	86.3	87.9	93.0	95.0	91.2	89.2	81.8	69.7	68.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	57.7	59.3	65.8	78.5	84.0	89.3	91.9	87.9	85.2	77.1	65.5	61.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	51.3	52.1	60.2	73.1	81.0	86.2	89.3	85.8	81.8	73.0	61.5	57.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	47.0	47.0	55.4	68.3	77.4	83.3	86.7	83.6	78.7	69.9	58.5	52.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	12.5	13.1	14.2	16.3	15.3	14.6	14.3	13.3	13.6	13.9	12.8	11.5
		MCDBR	20.6	21.1	22.5	23.6	19.3	17.6	17.5	16.2	16.8	17.4	17.3	16.8
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.316	0.322	0.365	0.403	0.448	0.511	0.521	0.515	0.426	0.364	0.328	0.312
	taud		2.446	2.420	2.356	2.277	2.186	2.053	2.038	2.040	2.265	2.443	2.501	2.503
	Ebn at Noon		270	285	282	277	266	248	244	241	258	266	264	264
	Edh at Noon		26	30	36	41	46	53	53	52	39	29	24	23
(44) All-Sky Solar Radiation	RadAvg		627	910	1196	1549	1732	1899	1912	1681	1385	1007	720	548
	RadStd		53	102	95	126	213	141	86	103	151	104	62	56

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (80 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+239	+140

Nomenclature: See separate page